AGRICULTURAL MANAGEMENT SYSTEM

A Java Full Stack Web Application for Farmers

Summer Internship (IT3711) | Fantasy Solution, Trichy

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Company Profile: Fantasy Solution



Trichy, Tamil Nadu

Software Development & Training

An IT services company delivering web applications, software solutions and technical training for students and fresh graduates.

Industry	IT Services & Software Development
Services	Web & mobile apps, database design, REST APIs, testing, training
Technologies	Java, Spring Boot, HTML, CSS, JavaScript, Bootstrap, MySQL
Tools	Eclipse, IntelliJ IDEA, VS Code, Maven, Git & GitHub

Introduction & Objectives

Why this project?

- Farmers rely on scattered, manual sources for crop, weather, irrigation, price and scheme data
- This causes delays, poor decisions and wasted resources
- Solution: one secure, centralized web platform

Objectives

- Secure registration and login
- Crop information management
- Weather and irrigation guidance
- Market prices and govt. schemes
- Better, timely farming decisions
- Apply Java Full Stack, REST APIs and MySQL

5	3-Tier	10+	30 Days
Core Modules	Architecture	REST Endpoints	Internship

Existing vs Proposed System

Existing: Manual Methods

- Crop choice by guesswork or tradition
- No localized, real-time weather
- Over / under irrigation, water wastage
- Market prices scattered across sources
- Low awareness of govt. schemes
- Missing or outdated records

Proposed: AgriFlow

- Season and soil based crop guidance
- Weather details by location
- Smart irrigation schedule and cost
- Centralized market price table
- All govt. schemes in one place
- MySQL records + admin dashboard

System Architecture & Block Diagram

Presentation Layer

HTML5, CSS3, JavaScript, Bootstrap

Business Logic Layer

Java Spring Boot, REST APIs, Maven

Database Layer

MySQL: users, crops, prices, schemes

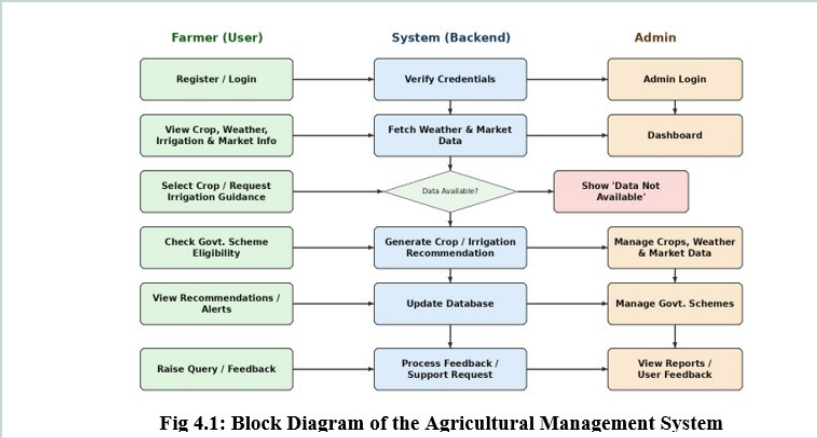


Fig: Block diagram of the system

Tools: Eclipse / IntelliJ / VS Code, Git & GitHub, Postman

Key Modules

Crop Management

Crop name, soil, season, location; admin add / update / delete

Weather

Temperature, humidity, rainfall, wind by location

Irrigation Guidance

Method, timing and frequency by crop and soil

Market Prices

Commodity, market, price and date details

Govt. Schemes

Scheme name, eligibility, benefits, how to apply

Admin Module: manage users, crops, prices and schemes from one dashboard

Core Workflows (Algorithms)

User Login

1. Enter email and password
2. Validate required fields
3. Request goes to AuthController
4. Check email exists in database
5. Verify password
6. Invalid: show error message
7. Valid: redirect to dashboard

User Registration

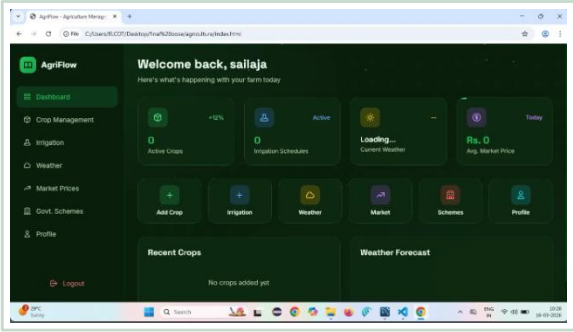
1. Enter name, email, phone, password
2. Validate all fields
3. Check if email already exists
4. Exists: reject request
5. Unique: create new user record
6. Store in MySQL database
7. Redirect to login page

Code Implementation

Module	Endpoint
Login	POST /api/auth/login
Register	POST /api/users/register
Crops	POST /api/crops/add, GET /api/crops/all
Weather	GET /api/weather/{location}
Irrigation	GET /api/irrigation/guidance
Market	GET /api/agriculture/market-prices
Schemes	GET /api/agriculture/schemes
Admin	GET /api/admin/users, DELETE /api/admin/crop/{id}

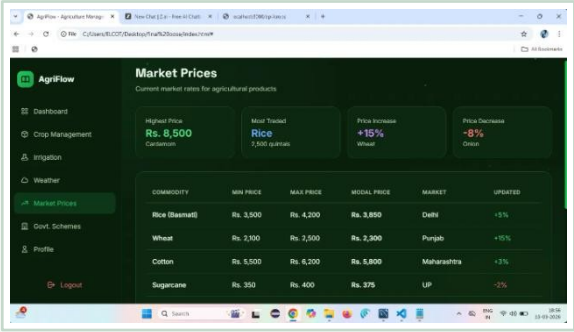
Controller > Service > Repository pattern with Spring Boot

Sample Output: AgriFlow Screens



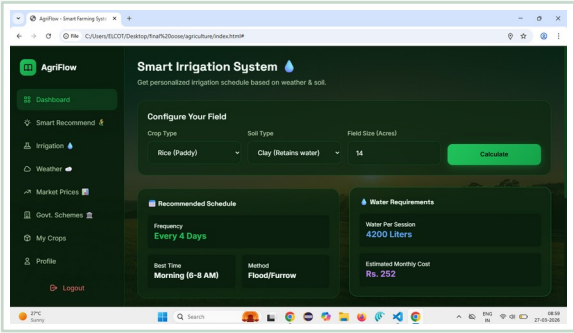
Farmer Dashboard

Crops, irrigation, weather and market price summary



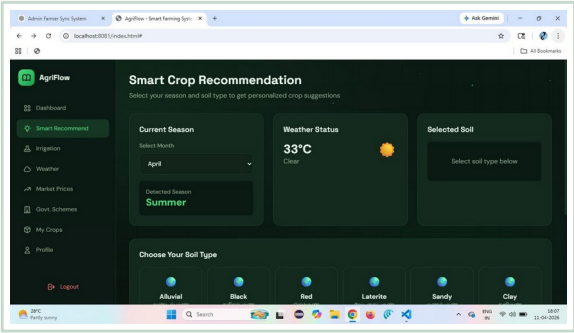
Market Prices

Min, max and modal prices by commodity and market



Smart Irrigation

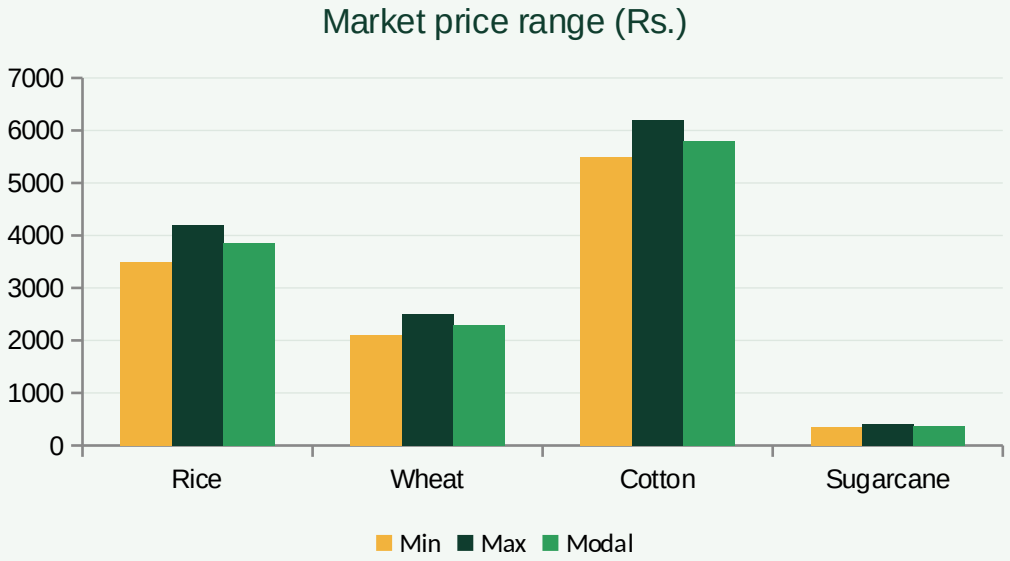
Schedule, water need and cost from crop, soil and acres

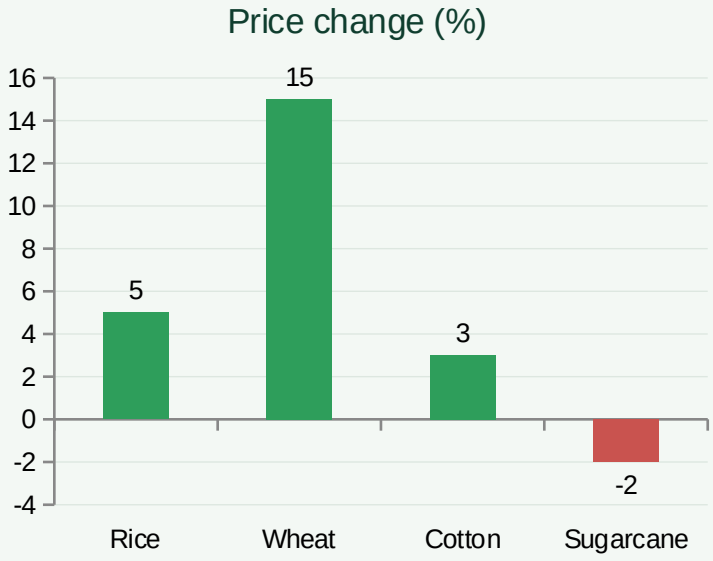


Crop Recommendation

Suggestions by season, weather and soil type

Analysis: Market Prices & Irrigation





Irrigation result: Rice (Paddy), Clay soil, 14 acres

Every 4 Days	4,200 L	Rs. 252	6-8 AM
Frequency	Water / session	Est. monthly cost	Best time

Conclusion & Future Work

Conclusion

- One platform for crop, weather, irrigation, price and scheme info
- Layered Spring Boot + Bootstrap + MySQL design
- Saves farmers time and improves resource use
- Hands-on Full Stack and REST API experience

Future Work

- Real-time weather and market price APIs
- AI crop recommendation and disease detection
- IoT soil sensors
- Android / iOS mobile app
- SMS and email alerts
- JWT security and cloud deployment

Thank you!